SIMARJOT SINGH ANAND

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CS undergraduate building end-to-end AI systems, from computer vision on edge hardware (patent filed) to LLM-powered search over company annual reports. Seeking AI/ML and software engineering internships.

EDUCATION

Vellore Institute of Technology, Chennai *2024 – 2028*
B.Tech, Computer Science & Engineering **CGPA: 8.3 / 10**
Class XII (CBSE): 93.0% · Class X (CBSE): 91.2%

TECHNICAL SKILLS

Languages	Python, C, C++, Java, SQL, JavaScript, PHP, R
AI / ML	scikit-learn, XGBoost, SHAP, SMOTE, YOLOv8, MobileNetV3, MediaPipe, OpenCV, ONNX Runtime, RAG, embeddings, Gemini API
Data & Backend	pandas, NumPy, DuckDB, MySQL, SQLite, ChromaDB, Flask, FastAPI, Streamlit
Web & Tools	HTML, CSS, React, Git, GitHub, pytest, Raspberry Pi, Kaggle

PROJECTS

Ambient Vibe Intelligence System – *Patent filed* *Raspberry Pi, YOLOv8, MobileNetV3, FastAPI*

- Co-invented an edge-AI IoT device that turns a Raspberry Pi camera feed into a real-time "vibe" score for hospitality venues by fusing lighting, occupancy, scene-aesthetic and infrastructure signals.
- Built the on-device inference pipeline: YOLOv8-nano for occupancy and MobileNetV3 for scene classification.
- Served the scores through a Flask/FastAPI REST API consumed by an artist–venue matching platform that connects restaurants with local musicians; filed as a formal patent disclosure through VIT Chennai, with a working end-to-end prototype at TRL-3.

Single-Frame Face Liveness Detection – *Team project* *Raspberry Pi 5, MediaPipe, MiniFASNet, ONNX, OpenCV*

- Built a face anti-spoofing system that tells a real face from a printed photo or a replayed video using one RGB frame from an ordinary camera, running entirely on a Raspberry Pi 5 with no depth sensor and no blink or head-turn challenge.
- Combined MediaPipe face and pose checks, two MiniFASNet models at different crop scales and an FFT cue for screen moiré, smoothed over 20 frames, with live verdicts streamed to a Flask dashboard.
- Reached 3.4% ACER on OULU-NPU Protocol 1 at 16.2 ms/frame on a laptop CPU; INT8 quantisation cost just 0.8 ACER.

FilingLens: AI Analyst for SEC Annual Reports – *In progress* · GitHub *Python, DuckDB, RAG, ChromaDB, Gemini*

- Building an AI analyst that answers plain-English questions about public companies by combining retrieval over 10-K report text with SQL over financial data.
- Wrote a data pipeline that loads 10-K filings, XBRL financial statements and daily stock prices from SEC EDGAR and Yahoo Finance into DuckDB, with every figure traceable to its source filing.
- Parsed 84 annual reports into 1,862 sections and 21,005 passages, embedded them with bge-base-en-v1.5 into ChromaDB, and built a RAG pipeline that answers with numbered citations; unit-tested with pytest.

Customer Churn Prediction – Live demo · GitHub *scikit-learn, XGBoost, SHAP, Streamlit*

- Built an end-to-end churn model on the IBM Telco dataset (7,043 customers, 26% churners), applying SMOTE to the training split only to handle class imbalance without data leakage.
- Compared Logistic Regression, Random Forest and XGBoost on recall and ROC-AUC rather than accuracy (best AUC 0.83), and explained the model with SHAP (short tenure and month-to-month contracts drive churn).

Mannsay: Online Counselling Platform – *Client project* · Live site *HTML, CSS, JavaScript, PHP, MySQL*

- Designed and delivered a responsive website for a clinical psychologist, with online appointment booking, client login and a wellness self-check.
- Built the PHP/MySQL backend for user accounts, appointments and progress tracking, plus a dashboard to manage appointment status and email notifications via EmailJS.

Campus Finder: Lost & Found System – *Team project* · GitHub *SQL, SQLite, Python, JavaScript*

- Co-built a campus lost-and-found app: a Python server over a six-table SQL schema with weak entities, which scores every lost–found pair out of 100 (category, location, date, description) in one SQL self-join.

ACHIEVEMENTS & CERTIFICATIONS

- Co-inventor on a patent disclosure filed through VIT Chennai (Ambient Vibe Intelligence System).
- Certifications: 19 Google Cloud Skills Boost courses (GDG on Campus VIT Chennai Study Jams, 2025); Infosys Springboard Basics of Python and Programming Fundamentals using Python (Feb 2025).
- Languages: English, Hindi, Punjabi.